

Data Visualisation Mini Project

Group 3 (Cheung Man Ting, Sze Hiu Laam, Yam Ching Long)

Global Unemployment Market Analysis

Introduction and Objectives

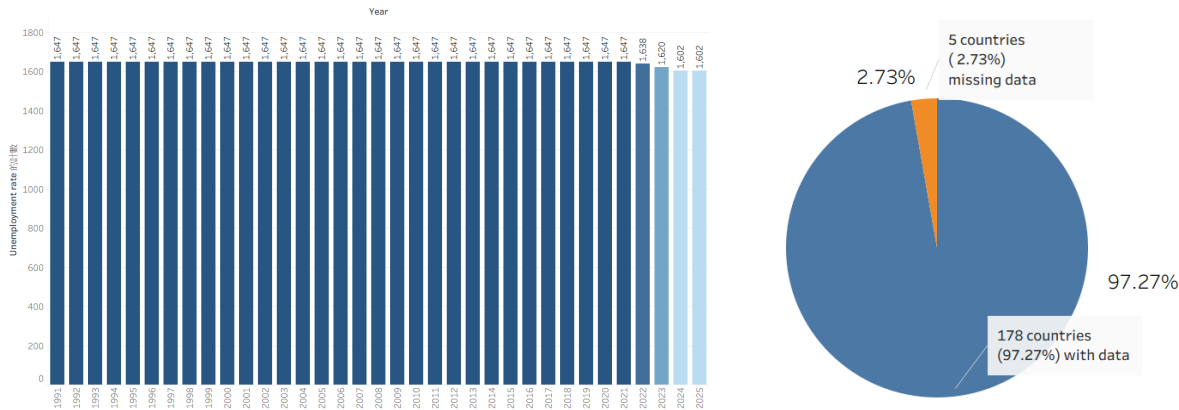
This project analyses the global unemployment patterns from 1991 to 2025 by focusing on the following 3 main objectives:

- **Macroeconomic Shocks:** Evaluating how major global economic events affect market resilience across different Human Development Index (HDI) tiers.
- **Gender Disparity:** Analysing the balance between female and male unemployment rates and how education influences this gap.
- **Generational Gaps:** Examining the employment differences between youth (ages 15–24) and adults (ages 25+) to identify which group bears a heavier market strain.

Dataset Description

The foundation of the dataset sourced from Kaggle and rests on six primary variables natively tracked across a **35-year period (1991–2025)** for all **183 countries**, providing the basic demographic dimensions (sex and age) for the analysis. To facilitate deeper contextual segmentation and uncover structural socioeconomic patterns, we enriched the baseline dataset by integrating three additional analytical dimensions, namely **Continent, Human Development Index, and Education Level**.

Data columns (total 9 columns):				
#	Column	Non-Null	Count	Dtype
0	iso_code	57519	non-null	object
1	country	57519	non-null	object
2	sex	57519	non-null	object
3	age	57519	non-null	object
4	year	57519	non-null	int64
5	Unemployment rate	57519	non-null	float64
6	continent	57519	non-null	object
7	Human Development Index	56889	non-null	object
8	Education Level	56574	non-null	object
dtypes: float64(1), int64(1), object(7)				

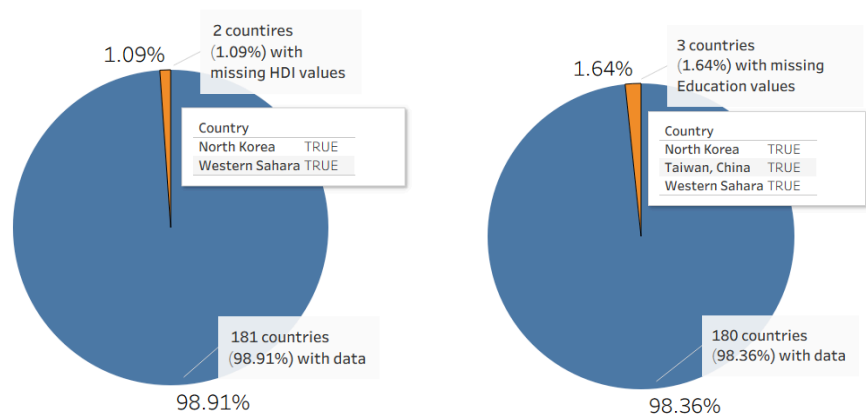


In terms of data completeness for the primary dataset, a total of 126 records (~0.2%) are missing during the **2022–2025 period**. Further investigation reveals that only 5 (2.7%) out of 183 countries exhibit missing values during this timeframe. Given this negligible proportion, a conscious decision was made to retain these nations to preserve the integrity of their historical data from 1991 to 2021. To address the resulting sample asymmetry, the **Median** was systematically utilized as the core metric. This methodological choice effectively mitigates the impact of data missingness and potential outliers, ensuring consistent and robust comparative trends across the Continent, Human Development Index, and Education Level.

Continent	Country	Year																	
		2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	
Africa	South Sudan	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9		
	Sudan	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9			
Asia	Lebanon	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9		
	Palestine (State of)	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9			
Europe	Ukraine	9	9	9	9	9	9	9	9	9	9	9	9	9					

Dataset Description (Cont'd)

Regarding the supplementary macro-dimensions, a minor cross-sectional gap persists, with **2 to 3 countries** lacking baseline data for Education Level and Human Development Index (HDI) classifications. To address the minor cross-sectional gaps in the supplementary macro-dimensions, listwise deletion was applied. The 2 to 3 countries lacking baseline classifications were excluded exclusively from the HDI and Education Level sub-analyses. Given that this omission accounts for **less than 2% of the global pool**, its impact on the aggregated baseline remains statistically negligible.



Methodology

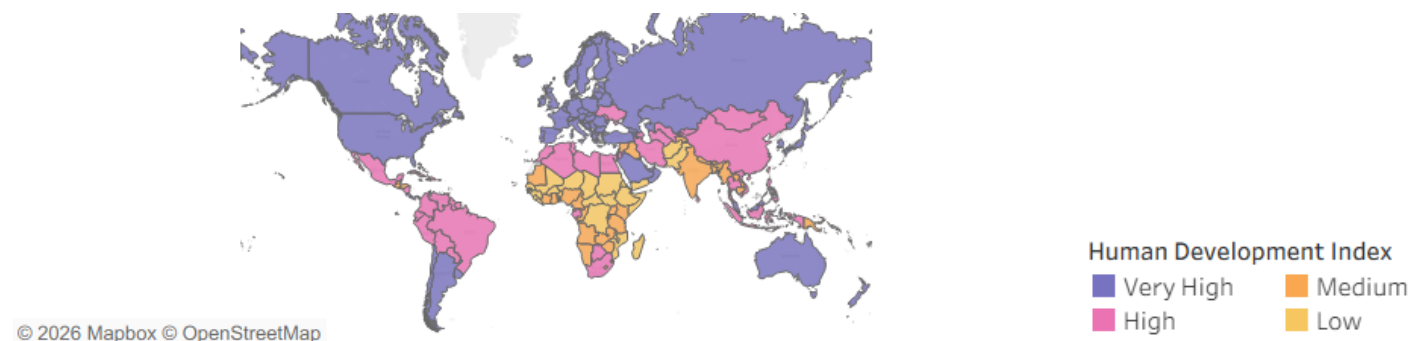
To analyse global unemployment patterns across our three core research objectives, we utilised **Tableau Desktop** to build an interactive, multi-dimensional analytical dashboard.

Tools	Descriptions
Word Cloud	Provide a visual representation of the five main continents
Map	Display the location of the continents and delineate regional boundaries to display the global spatial distribution of the Human Development Index (HDI) and educational attainment levels
Line Graph	Demonstrate the trends of the unemployment rate
Categorical Dot Plots	Demonstrate individual country-level data points vertically to evaluate the distribution and outliers of the unemployment rate across different HDI categories
Pie Chart	Illustrates the proportional composition of the HDI levels and the education levels
Scatter Plots	Analyse the correlation of the unemployment rates between different genders
Positive-Negative Horizontal Bar Chart	Demonstrate the trends of the unemployment rate of females over males across the four education tiers
Vertical Grouped Bar Chart	Demonstrate a side-by-side comparison between the unemployment of the youth and adults across the four HDI development tiers
Area Graph	Illustrate the unemployment disparities of youth and adults in different continents
Filters / Highlighters	Enable users to isolate specific cohorts to examine localised anomalies, cultural shift, gender gap and etc without changing the underlying dashboard architecture
Slider	Enable users to animate the dashboard chronologically, seamlessly tracking yearly fluctuations and observing the immediate “shock-wave effect” of major macroeconomic crises.

Highlights

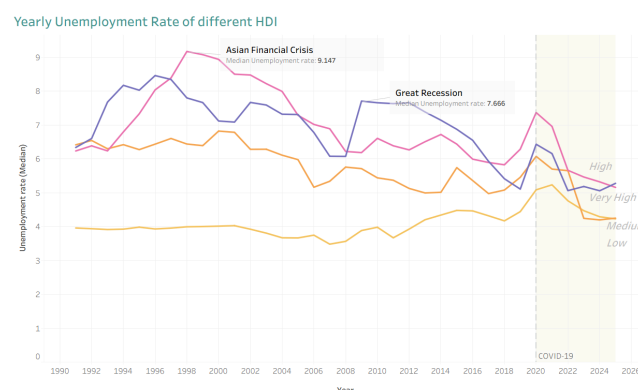
1. Macro-Economic Resilience

The analysis evaluates macroeconomic shocks and market resilience through the lens of the Human Development Index (HDI). The four tiers: Very High, High, Medium, and Low, are geographically distributed across different global regions as below.



Identification of Global Spikes (1991–2025)

By leveraging a 35-year historical timeline, the visualization clearly isolates three major macroeconomic shocks that broke long-term stabilization patterns and triggered severe unemployment spikes. Specifically, during the 1998 Asian Financial Crisis, countries with "High" HDI were hit the hardest. Next, during the 2009 Great Recession, the shock heavily impacted the "Very High" HDI group. Later, the 2020 COVID-19 Pandemic disrupted the entire world, breaking the usual safety of less-developed nations and causing sudden spikes across all development levels at the same time.

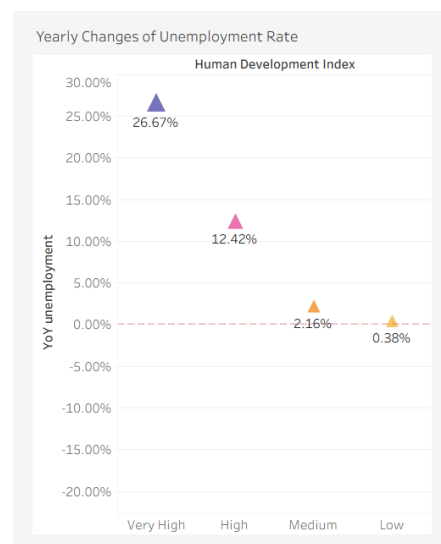


Shock Severity Across Development Tiers

The above time-series chart reveals that higher development tiers, namely "Very High" (Purple) and "High" (Pink), exhibit significantly higher sensitivity to cross-border economic crises. In contrast, "Low" HDI (Yellow) nations show a flat, steady line, meaning they stayed mostly independent of global market shocks.

To see the true speed and strength of these shocks, we can look at the Yearly Changes of Unemployment Rate chart. This chart highlights the year 2009 with the biggest gap in economic resilience:

- The "Very High" HDI tier suffered a massive 26.7% yearly increase in unemployment.
- The "High" HDI tier followed with a 12.4% yearly jump.
- In sharp contrast, the "Low" HDI tier was barely affected, changing by a tiny 0.4%.



Highlights (Cont'd)

1. Macro-Economic Resilience (Cont'd)

Shifting Heavy-Impact Zones and Outliers

By moving from general averages to country-level data points, we can pinpoint exactly where the economic damage was worst:



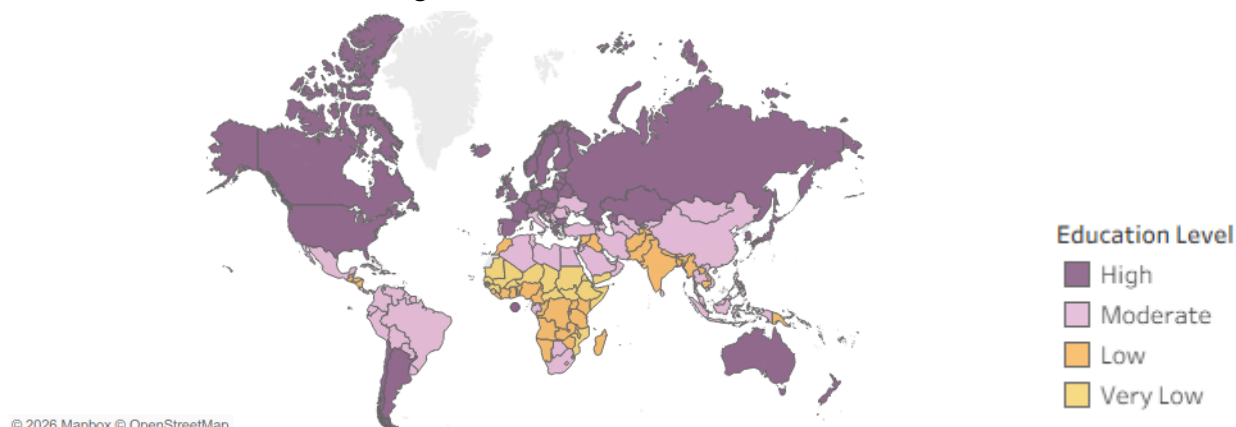
The 2009 Great Recession: The shock hit the "Very High" HDI tier severely. North Macedonia (Europe) stood out as an extreme case, facing a devastating median unemployment rate of 32.35%.

The 2020 COVID-19 Pandemic: The economic damage shifted to new areas, dragging down both "Very High" and "High" HDI tiers. This time, Armenia (Asia) stood out as a major high-unemployment outlier on the chart.

In conclusion, the charts prove highly developed economies (High and Very High HDI groups) face much larger jumps in unemployment and heavier labor market strain during global crises. While less-developed nations stay mostly safe because they are less connected to global markets, advanced economies suffer from a rapid "shock-wave effect." This happens because deep ties to international trade and global financial networks instantly turn into pathways that spread high unemployment when a global crisis hits.

2. Structural Gender Parity

The education level is divided into four categories: High, Moderate, Low and Very Low. We first plot a world map showing the education level of different regions:



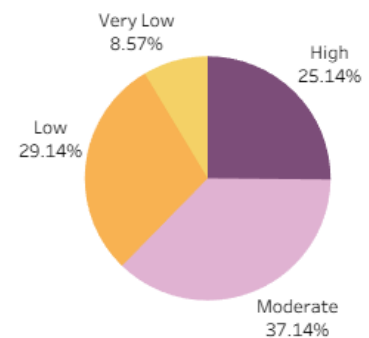
Each colour represents an education level tier, see the legend above.

Highlights (Cont'd)

2. Structural Gender Parity (Cont'd)

In addition, we plot a pie chart showing the distribution of the education level as shown:

From the pie chart, we can see that the proportion of countries at Moderate education level is the largest, which constitutes about 37% of the whole population. The second largest is Low education level, which constitutes about 29%, followed by High education level and Very Low education level.



Correlation between Male and Female Unemployment Rate

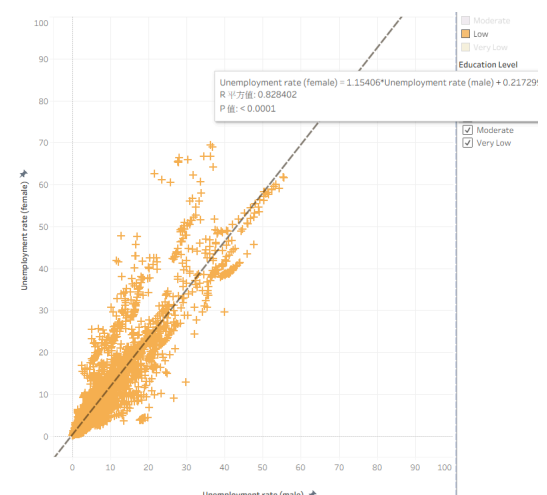
Furthermore, we plot a scatter plot which demonstrates the relationship between male and female unemployment rate:

Same as the world map, the different colours on the scatter plot represent different education levels. For each education level, a trend line was plotted to reveal the trend between male and female unemployment rate.



We discovered that, among the four education levels, the trend line at Low education level has the greatest slope which is about 1.154:

This means that countries at Low education level have their data points shifted the most towards the left among all education levels. It implies that the female unemployment rate is higher than that of male. In these countries, many females need to work for necessity, and they may engage in black-market jobs such as prostitution. These females are not counted in the employment population. As a result, the female unemployment rate is seemingly higher than that of male.

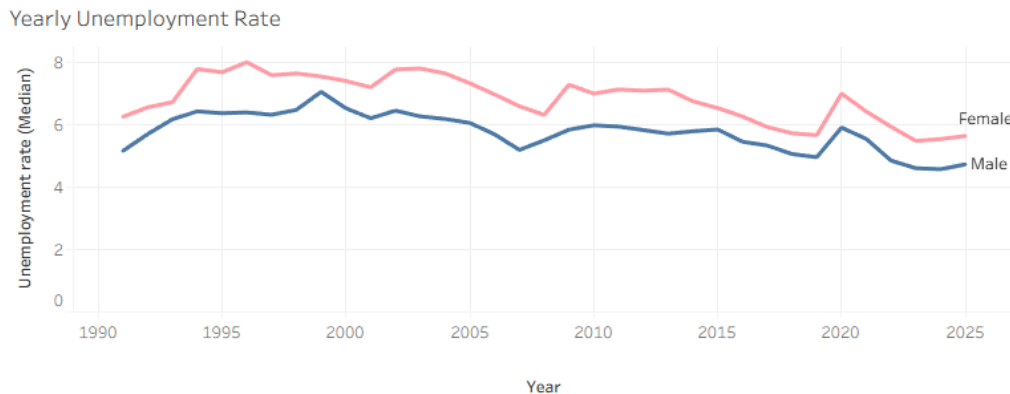


Highlights (Cont'd)

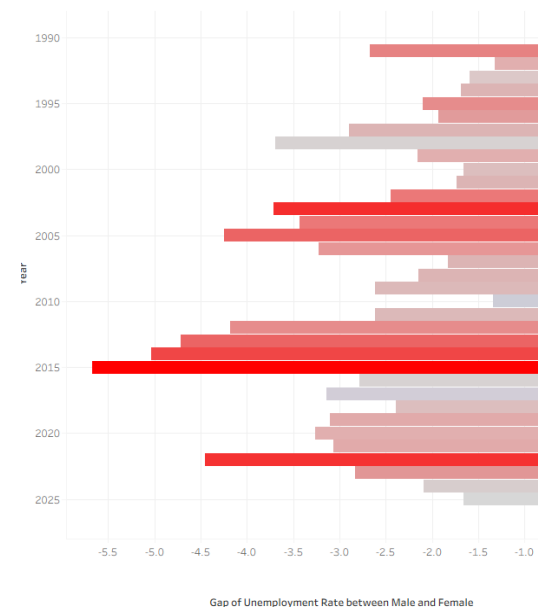
2. Structural Gender Parity (Cont'd)

Year-over-Year Male and Female Unemployment Rate

We then plotted a line graph showing the global unemployment rate of male and female over the years:

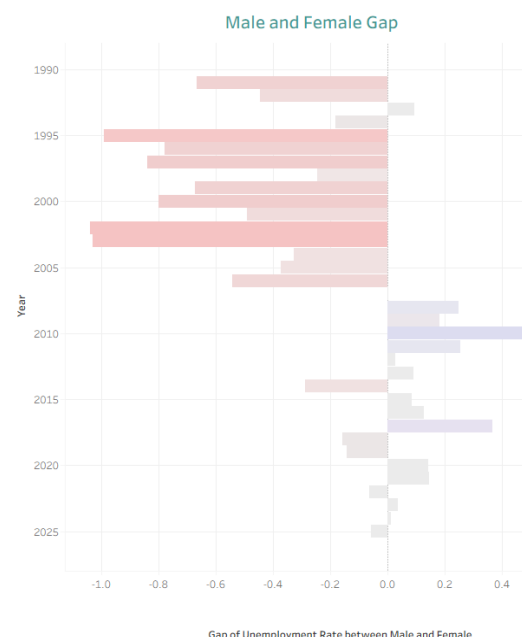


Obviously, the female unemployment rate has always been higher than that of male over the past years. On the other hand, we plotted a horizontal bar chart which shows the unemployment gap between male and female over time: Here, unemployment gap = male unemployment rate - female unemployment rate. If it is a positive number, then male unemployment rate is higher than that of female. Conversely, if it is a negative number, then female unemployment rate is higher than that of male.



We then filtered the High education level to see how the unemployment gap in this category changes with year:

It can be seen that the gap is becoming smaller and smaller in recent years. This is probably due to the rise of feminism in western countries in recent years, which makes people attach importance to gender equality in job opportunities. Hence, there are more job openings for females.



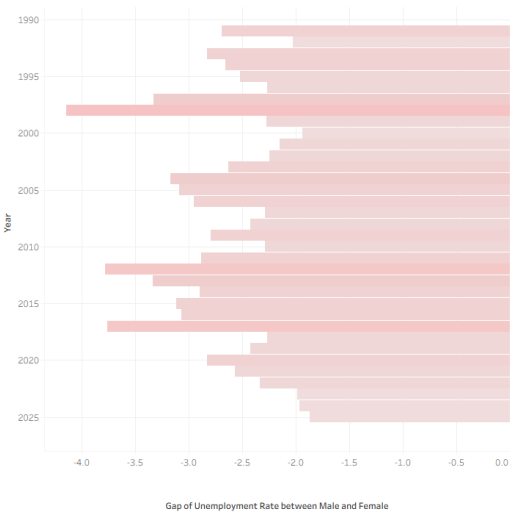
Highlights (Cont'd)

2. Structural Gender Parity (Cont'd)

Year-over-Year Male and Female Unemployment Rate (Cont'd)

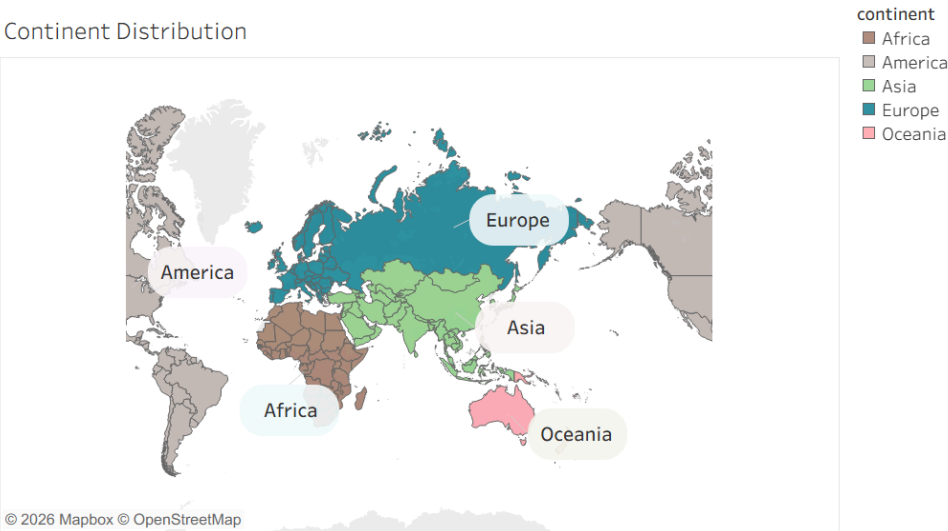
The second thing we want to investigate is the unemployment gap of countries at Moderate education level:

It is evident that the female unemployment rate is always higher than that of male over the past years. In these countries, females are not expected to work but to be housewives. The countries tend to protect males' careers for fear that females would steal their jobs. Sexism remains severe in these countries.

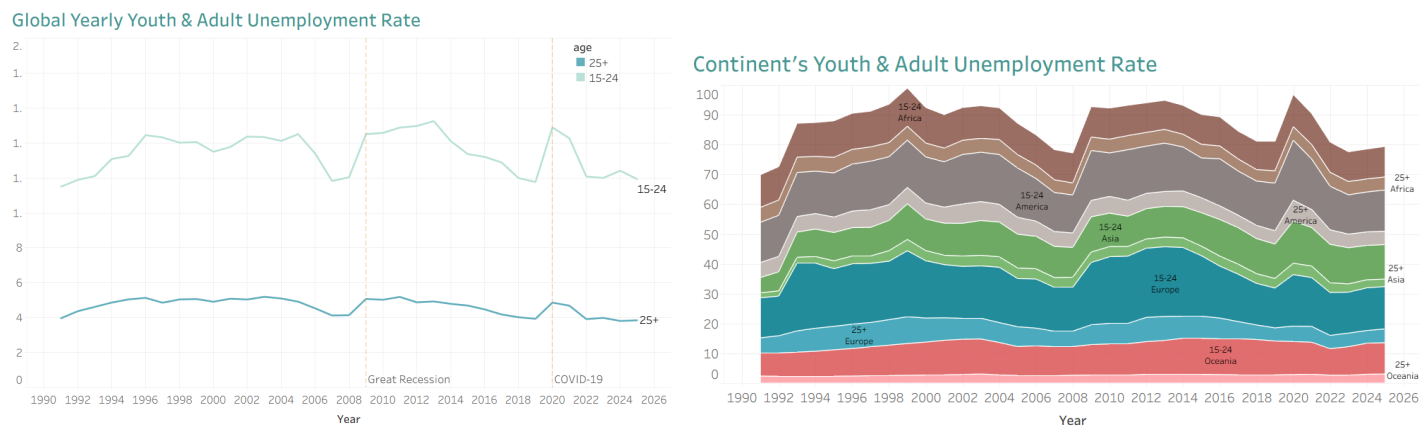


3. Generational Workforce Disparity

Another world map is created to analyse data across the five primary regions: Africa, America, Asia, Europe and Oceania.



Overview of the Unemployment Rate of the Youth and the Adults



The unemployment rate of the youth has consistently outpaced that of the adults throughout the years. When macroeconomic shocks occur (e.g., the Great Recession, COVID-19, etc.), the younger are proven to be significantly more vulnerable to labour market volatility.

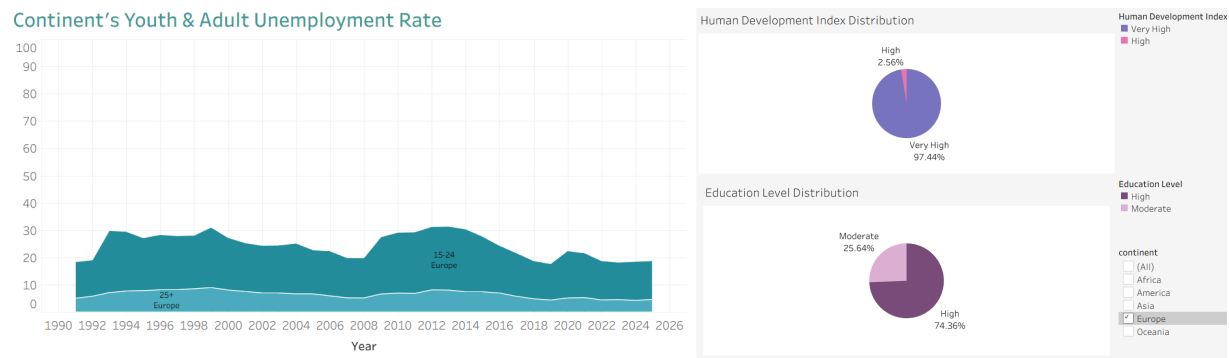
Highlights (Cont'd)

3. Generational Workforce Disparity (Cont'd)

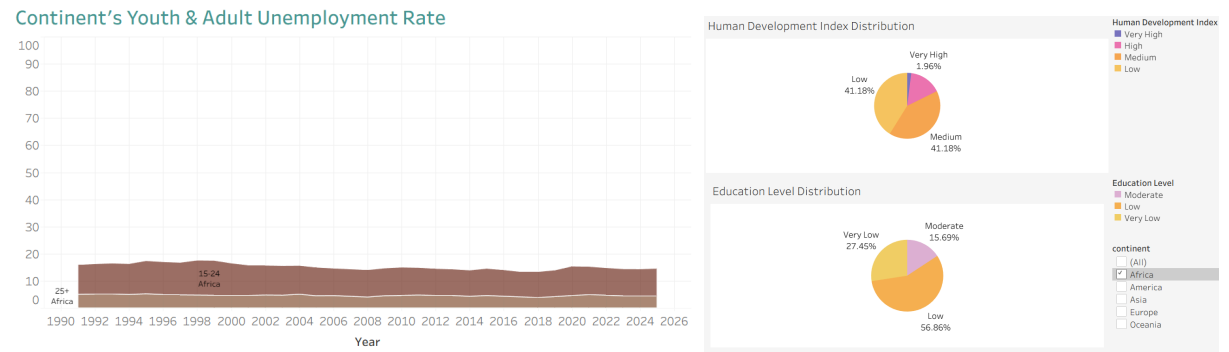
Overview of the Unemployment Rate of the Youth and the Adults (Cont'd)

The area graph shows the unemployment rate of the youth and the adults in different continents. Darker hues represent the youth population within a continent, while the lighter shades represent adults. Obviously, the youth in America (dark grey) and in Europe (dark blue) encounter the most severe job-hunting obstacles.

Generational Differences in Continents with Different Human Development Index Levels and Education Levels



The unemployment rate of the younger generation is vastly higher than that of the mature workers in the highly developed and highly educated region. In Europe, 100% of the countries fall into “High” or “Very High” Human Development Index (HDI), and over 70% feature highly educated populations (please see the first two pie charts above). Yet, 20-30% of the youth in Europe are unemployed.



On the contrary, only 10-20% of the youth in Africa remain unemployed, where less than 20% of the countries in Africa belong to the “High” or “Very high” developed or educated brackets. The unemployment rate of the youth in Africa has exhibited much lower volatility over the years when compared to Europe.

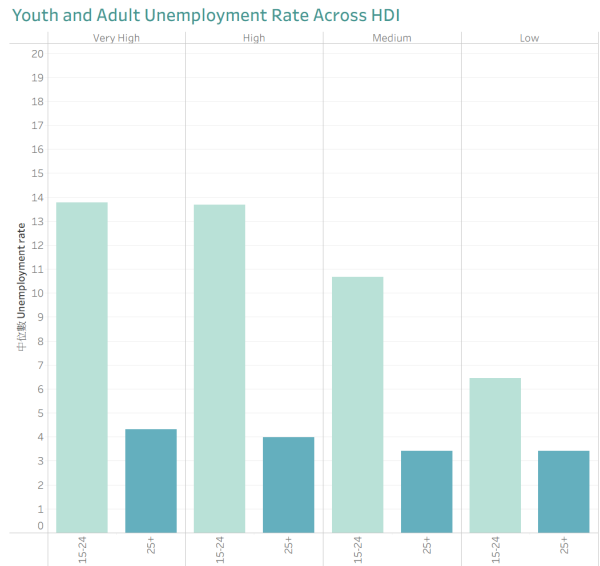
Highlights (Cont'd)

3. Generational Workforce Disparity (Cont'd)

Higher Human Development Index, More Fragile the youth

In accordance with the observations we have gained above, the bar charts clearly demonstrate that a higher HDI correlates with both a higher youth unemployment rate and a wider generational disparity gap.

A plausible explanation is that the developed region possesses highly specialised, knowledge-based industries with steeper entry barriers for the youth, who have less practical experience and a less robust educational background, to secure a place. In contrast, less developed economies may rely on informal sectors, coupled with lower legal working age, allowing the youth to enter the workforce much earlier and more fluidly.



Effectiveness

- **Facilitating Hypothesis Generation through Layered Dimensions:** The main strength of this Tableau dashboard is how it splits the unemployment data by both Time (1991–2025) and Development Level (HDI Tiers). Seeing that the "Very High" HDI had much larger jumps during crisis years helped our team form a key idea: "countries that are deeply connected to global trade and finance are actually the most vulnerable to sudden job losses when a global crisis hits."
- **Connecting Time and Maps for Clear Storytelling:** By linking a Dynamic Year Filter to both the time-series chart and supporting charts at the same time, our dashboard makes static figures tell a clear historical story. This interactive setup allowed us to easily see how job losses moved across the world over time.

Challenges

We encountered several challenges in this project. The first one is that the first dataset we found, which is the unemployment dataset, only contains 5 features. Only a few graphs could be plotted due to the limited number of features. In view of this, we have found other two datasets: HDI (Human Development Index) and Education Level. We then joined them with the unemployment dataset so there are more features to analyze. The second challenge is that we used countries for analysis at first but later found that it was not a good option. We plotted a line graph showing the change of unemployment rate of different countries over time, and we only chose the "representative" countries for each continent to show on the graph because there are numerous countries. One of our groupmates questioned what makes the countries "representative" for that continent? Is it in terms of the size or the status? She suggested that we should consider clustering the countries into continents so each continent can represent the many countries in it. We finally adopted her suggestion.

Limitations & Future Work

After grouping the countries into continents, we wanted to perform the analysis using the mean of the unemployment rate of each continent. Unfortunately, we cannot just sum up the unemployment rate of each country then take the average of it, because the definition of the unemployment rate is the unemployment population of that country divided by the labour force of that country. Adding up the unemployment rate of each country is meaningless. After serious consideration, we simply used the median of the unemployment rate for each continent in our analysis. The downside is that it may not reflect the true unemployment situation of that continent.

The HDI (Human Development Index) Tier (Very High, High, Medium, Low) was from the year 2023. For simplicity, we combined this dataset with the unemployment dataset which spans across 1991 - 2025. The more accurate way is to find a dataset which shows the HDI Tier of each country over the past years. If such a dataset is found, we can conduct some intriguing analysis such as the dependence of unemployment rate on the HDI of a country. Besides, we didn't handle the missing values of the HDI Tier and education level. We just left them blank. If time allows, we can utilize some machine learning algorithms like Random Forest and Support Vector Machine(SVM) to replace the missing values with categorical values.

Conclusion

While advanced economic development and high educational attainment are traditionally viewed as benchmarks of societal progress, this analysis demonstrates that they do not guarantee a more stable or equitable labour market. On a macro scale, highly developed nations bear the brunt of cross-border economic crises. Their deep integration into global trade and financial networks exposes them to severe unemployment spikes during global downturns. In contrast, less developed nations remain largely insulated from these volatile shocks due to their relative detachment from the global financial system. Apart from macroeconomic vulnerability, advanced economies possess highly specialised, value-added industries that may create steeper entry barriers for the younger generation. Despite these, high development yields benefits and improves gender parity. The unemployment gap between males and females has steadily narrowed in the higher education tiers, which may imply that higher educational attainment empowers women to secure positions more easily, and the organisations in the more educated ecosystem may also be more committed to providing equal professional opportunities for both genders.

Reference

- (Main) Global Employment Unemployment Rates 1991-2025 [\(Link\)](#)
- (Continent) Country Mapping [\(Link\)](#)
- Human Development Index (HDI) by Country [\(Link\)](#)
- Countries Education Level and Income [\(Link\)](#)

Distribution of Work

Cheung Man Ting	Sze Hiu Laam	Yam Ching Long
• Dataset Sourcing (Edu Level) • Data Characteristics • Dashboard Presentation • Insights (Macro-economic)	• Dataset Sourcing (HDI) • Graph Plotting & Methodology • Insights (Generation) • Conclusion	• Dataset Sourcing (Main data) • Data Cleansing • Graph Plotting • Insights (Gender) • Challenges & Future Work